ISEL | DEI | AMD / MLDM

Machine Learning and Data Mining

1. Context and objective (*just read; do nothing yet!*)

This MoP starts from a provided analytical dataset of the Palmer Penguins observations. It continues the learning path from data preparation to predictive modelling.

The activity comprises **three connected notebooks**. By completing this MoP, you should be able to:

- **construct feature representations** given the information available at prediction time
- **train, evaluate, compare** model configurations under controlled experimental conditions
- **select, save, reuse** a trained configuration to predict the target of new observations

The target throughout the activity is `species`. The emphasis is on understanding the decisions linking these stages, not merely obtaining high classification accuracy.

2. Preparation and working principles

The starting point is the provided file:

- `01_input/penguins_analytical_dataset.csv` (*do not copy it from previous MoP*)

You will use Python/Jupyter, pandas, scikit-learn, joblib and Orange Data Mining (DM). Follow the notebook instructions and use the provided MoP workspace folders.

Recall: files whose names begin with `_mLDm_` are course originals and must remain unchanged, so:

- duplicate the three provided notebooks,
- remove the `_mLDm_` prefix from each working copy, and
- work only on those copies.

While completing the activity:

- consider prediction-time information when choosing attributes and representations
- keep the technical identifier `ID` separate from model features
- use expected-result checkpoints to verify work, not to replace (your own) interpretation
- compare configurations using the prescribed experimental controls

Distinguish:

- `fit()` to learn parameters
- `transform()` to apply transformations, and
- `predict()` to generate predictions

Complete `TODO` sections and answer questions, with concise, evidence-based explanations.

3. Files and execution order

Run the working notebooks in this order:

- `02_code/01_penguins_feature_representation.ipynb`
- `02_code/02_penguins_modelling.ipynb`
- `02_code/03_penguins_selection_deployment.ipynb`

01 reads the supplied analytical dataset and writes modelling datasets (`.csv` and `.tab`) and fitted feature transformers (`.joblib`) to `03_output`.

02 uses those datasets to compare feature representations, prediction scenarios, classifiers and scaling methods. Its experimental results inform the selection made in **03**.

03 uses the selected scenario's modelling dataset and saved feature transformer, trains and saves the final classifier (and scaler, when required), and processes new observations.

Complete each notebook before moving to the next. Preserve the generated modelling and deployment artefacts in `03_output`. They are required for subsequent operations, and the complete folder is included in the submission ZIP.

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4. Notebook 01 — Feature Representation

Open `01_penguins_feature_representation.ipynb`.

Starting from the same analytical dataset, construct representations for five prediction contexts:

- **A — Historical Field Study**, all available attributes
- **B — Anywhere in Antarctica**, without observation location
- **C — Whenever Needed**, without historical observation year
- **D — Biological Profile**, without geographic or temporal context
- **E — Morphological Measurements**, only the four morphological measurements

Select the attributes appropriate to each context. Use `ColumnTransformer` and `OneHotEncoder` to prepare numerical and nominal attributes, and construct the additional `A_noOHE` representation retaining the original nominal values.

Complete the transformations, validate the six modelling datasets, export their `CSV` and `TAB` versions and save the fitted transformers for all scenarios (A to E) with `joblib`.

Check: the six representations preserve same observations and target; the `ID` supports correspondence across scenarios (is not a learning feature); saved transformers will be used later.

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5. Notebook 02 — Modelling

Open `02_penguins_modelling.ipynb` (only after completing Notebook 01).

OrangeDM — (some) exploratory comparisons:

- compare DecisionTree trained with and without One-Hot Encoding (OHE) for scenario A; save `02_code/penguins_OHE_comparison.ows`

- compare the five prediction scenarios using DecisionTree, kNN and Naive Bayes; save `02_code/penguins_learners_comparison.ows`
- inspect model structures, accuracy and classification errors; consider why independently sampled train/test partitions limit comparisons across separate OrangeDM evaluations

Python/scikit-learn — controlled comparisons:

- define one reproducible, stratified 80/20 holdout using observation IDs, so every scenario uses the same training and test observations
- train, predict and evaluate DecisionTree, kNN and Gaussian-NaiveBayes classifiers
- compare original, min-max and standard-scaled features; fit each scaler **only on training data**, then apply it to training and test data
- summarize and interpret **45 configurations** (5 scenarios × 3 classifiers × 3 scaling methods)

Note: The alternative `A_noOHE` representation is used in OrangeDM, but not in the selected scikit-learn experiments.

Check: the Python experiments use identical train/test observations and consistent settings. Answer the notebook's reflection questions on feature representation, classifier performance, scaling, experimental control, and evaluation reliability.

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6. Notebook 03 — Model Selection and Deployment

Open `03_penguins_selection_deployment.ipynb` (only after completing Notebook 02).

Preliminary selection

Use your results to choose and justify one configuration actually evaluated in Notebook 02:

- prediction scenario
- classifier
- classifier hyperparameters, and
- scaling method

Exclude configurations that achieved 100% test accuracy. This is an educational constraint, **not** evidence of overfitting. The objective is to explore deployment operations with a reasonable configuration, not to establish the best model rigorously.

Final training and persistence

Recover the selected modelling dataset, the saved feature transformer and its input/output feature specifications.

Fit any required scaler and train the selected classifier on **all 333 observations**. Save the trained components needed for later prediction.

Deployment simulation

In a separate *simulated usage session*, reload the persisted components and process new observations from a CSV file. Validate required attributes and missing values, restore the

expected attribute order, apply the saved transformations, obtain predictions and save:

- `03_output/__deploy_penguins_predictions.csv`

Do not refit the feature transformer, scaler or classifier when processing new observations. Additional validation of values, data types or unknown categories is outside the scope of this MoP.

Complete the complementary `FunctionTransformer` exercise to examine identity transformations and their persistence. It does not change the main deployment implementation.

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7. Required notebook completion

Before submission, verify that:

- all three working notebooks complete their required tasks and checkpoints
- all `TODO` sections and requested written answers are complete
- the two OrangeDM workflows have been created and saved
- the selected configuration is justified and respects the selection constraint
- the deployment simulation reloads trained components and saves predictions for all new observations

Answers should be concise and supported by the observations and experimental results obtained during the activity. No separate report or screenshots are required.

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8. Submission checklist

Submit the following **three completed notebooks** and **two OrangeDM workflows**:

- `02_code/01_penguins_feature_representation.ipynb`
- `02_code/02_penguins_modelling.ipynb`
- `02_code/03_penguins_selection_deployment.ipynb`
- `02_code/penguins_OHE_comparison.ows`
- `02_code/penguins_learners_comparison.ows`

The submission must also include the **20 required output files** produced by the notebooks:

- **12 modelling datasets:** six feature representations (A–E and A_noOHE), each exported in CSV and TAB formats
- **5 feature transformers:** fitted and saved as JOBLIB files for scenarios A–E
- **1 final classifier:** `_deploy_penguins_model.joblib`
- **2 deployment CSV files:** `__deploy_penguins_new_observations.csv` and `__deploy_penguins_predictions.csv`

If the selected configuration uses scaling, the corresponding `_deploy_penguins_scaler.joblib` must also be present.

Use `_mLDm_xx_build_submission_zip.py` to validate the required artefacts and create the submission ZIP. The tool includes the complete `03_output` and `04_report` folders.

Before submission, verify that `04_report/submission_check.txt` reports successful validation and that the ZIP contains the expected artefacts.